

ADR 005 - Fix incompatibility with NetworkMessage key 'CorrelationId' to achieve OPC compliance

Status	PROPOSED
Date	Jan 14, 2026
Deciders	@Konrad Heidrich , @Thomas Weinschenk , @David Weiss , @Blumenstock, Werner (DI FA DSP SEN TRS)
Consulted	
Informed	OPCF

Context and Problem Statement

The `CorrelationId` key currently used in the NetworkMessage (s. [9.2.1](#)) is not compatible with the OPC UA specification Part 14. At the time the OI4-OEC-GL was developed, there was no equivalent for the meaning of the `CorrelationId` key in the OPC UA specification.

With the introduction of the OPC UA specification [Part 14 V1.05.04](#), additional keys were defined which might be able to replace the `CorrelationId` key.

Decision Drivers

- The OEC-GL describes the practical application of OPC UA Part 14 without implementing it in a compliant manner.
- The OI4 stands for the use of existing standards instead of reinventing the wheel.
- A standard-compliant description of the message bus payload will hopefully enable our users to reuse existing software components in the future.

Considered Options

In case the `CorrelationId` is not empty, it contains the `MessageId` of the related message which triggered these message. In many cases, this might be the “receiver” for this message and therefore the `ReplyTo` key, which is part of the [NetworkMessage](#), seems to fit perfectly in this case but not perfectly in case there is just a correlation between them.

Unfortunately, the OPCF does not explain exactly, how to use this key. Indeed, in chapter [NetworkMessageContentMask](#) it is explained as “Not used.”.

ReplyTo contains the **MessageId** to which the reply is based. This does not mean that this message is only intended for the recipient named in **ReplyTo**.

NOTE (new key) **ReplyTo** = (old key) **CorrelationId** = (value) **<MessageId>**

The **ReplyTo** key is mandatory in the OEC context. Therefore, the corresponding flags must be set in [NetworkMessageContentMask](#) (UA Part 14: PubSub - 6.3.2.1.1).

For better transperence the OEC guideline shall contain default values of all ContentMasks registers. This will make it much easier to implement a consistent “OEC interpreter”.

NOTE A good way to evaluate the different formats, depending on the ContentMask flags is to “play” around with the tool [OPC UA PubSub Formatter](#).

Decision Outcome/ Chosen Option

ReplyTo appears to be a valid substitute for **CorrelationId** but we should address the use of the field, marked as 'not in use' by the OPCF, at the OCPF.

Consequences/ Pros and Cons

- ➖ Not backward compatible to OEC-GL v1.1.x. Not only but also because Part 14 V1.05.x is not backward compatible to v1.04.x
- ➖ Existing OEC solutions are not able to get a new compliance batch, based on v2.x of the OEC-GL, without changes.
- ➕ Compatible to newest Part 14 specifiaction, which might make it easier to use open source components (The OI4 stands for the use of existing standards instead of reinventing the wheel).
- ➕ Consuming data from or providing data for existing OPC UA implementations should be easier

More Information

See [UA Part 14: PubSub](#) V1.05.04 from 29th of November 2024 for full information. Especially the following chapters:

- [UA Part 14: PubSub - 7.2.5.4 DataSetMessage](#)
- [UA Part 14: PubSub - 6.3.2.3.1 DataSetMessageContentMask](#)

- [UA Part 14: PubSub - 7.2.5.3 NetworkMessage containing DataSetMessages](#)
- [UA Part 14: PubSub - 6.3.2.1.1 NetworkMessageContentMask](#)

For a better understanding the tool [OPC UA PubSub Formatter](#) is recommended.